Indian Institute of Technology Kanpur

Department of Mathematics and Statistics

Introduction to Bayesian Analysis (MTH422)

Quiz 1, Date: February 3, 2025, Monday

Time: 8:00–8:40 AM (8:00–8:55 AM for DAP students) Spring 2025 Max point: 10

The answers need to be explicit. Writing some philosophical answer and claiming that it is correct will lead to no point. Calculation mistakes will be penalized.

1. Suppose X_1 and X_2 are two dependent random variables following $X_1 \sim \text{Normal}(1, 1)$ and $X_2|X_1 \sim \text{Normal}(1 + \rho(X_1 - 1), (1 - \rho^2))$. Find the marginal density function of X_2. The steps should be clearly written. Using results on convolution-based means and variances is not allowed. You can use the standard results of location-scale transformations of a normal distribution if necessary. (2.5 points)

2. Expert knowledge dictates that a parameter must be positive, and its prior distribution should have the mean 2 and variance 3. Is it possible to choose an exponential distribution as a prior choice here? If possible, write down the parameter of that exponential distribution. If not, propose a prior that satisfies the desired prior mean and variance. Justify. (2 points)

3. IIT Kanpur is going to launch its new website. To check the reach and modify it accordingly, the committee plans to record the number of visits before the launch for 30 days and also for 30 days after the launch. Suppose the click rate per hour for the old and new websites are θ_1 and θ_2, respectively. Besides, the total numbers of observed clicks are Y_1 and Y_2, respectively. Write down how you will decide whether the new website provides a better reach (in a Bayesian framework). Choose conjugate priors for θ_1 and θ_2 if they exist. If you cannot provide an analytical expression, provide a pseudo code. (2.5 points)

4. You want to understand the success probability of shooting a balloon at a fair in Kanpur. Suppose you denote the success probability by θ. Unlike a setup like many balloons, one balloon is placed on the shooting board at a time and the same target point. When you hit the first time, you stop and suppose Y denotes the total number of tries. Choose a model for Y appropriately and draw a posterior inference about θ. Choose a conjugate prior for θ (if it exists) and obtain the posterior mean, variance, and 95% equal-tailed credible interval for θ. If it doesn't exist, write a pseudo-code to obtain the posterior inference about θ. (3 points)